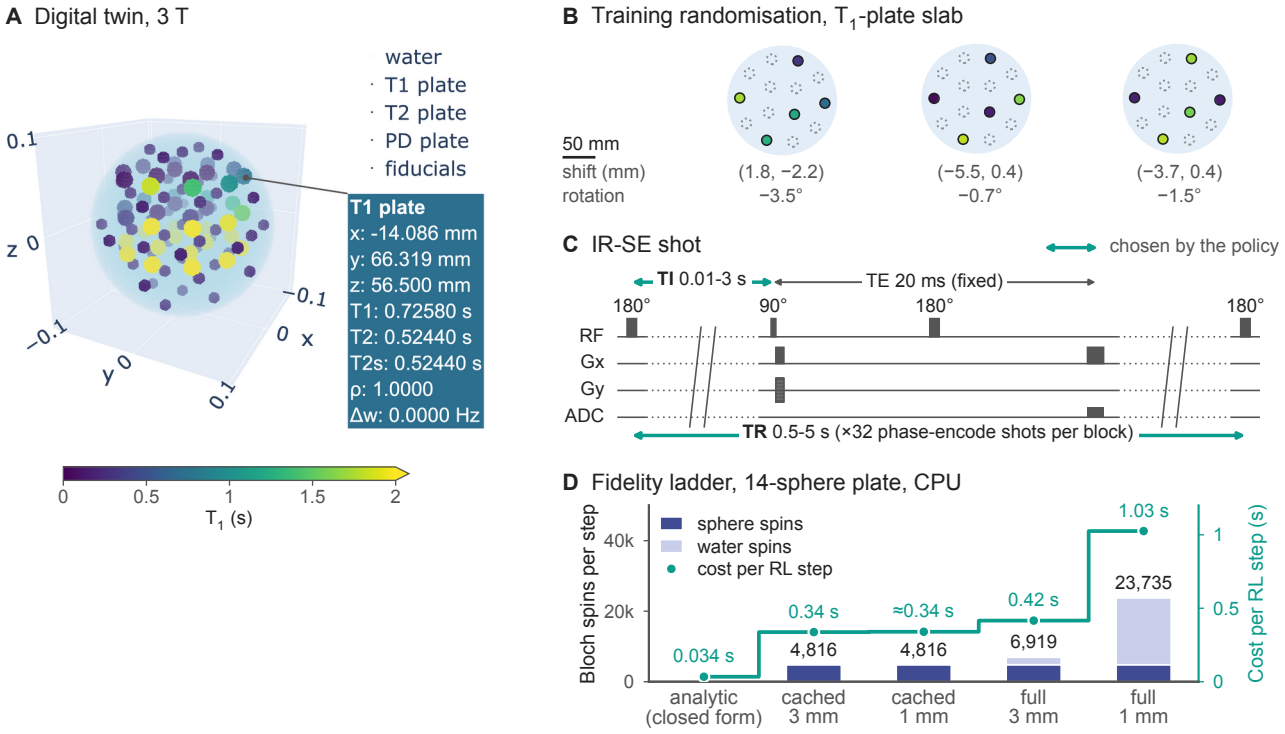

A Digital twin, 3 T
water
T1 plate
T2 plate
PD plate
fiducials
T1 plate
x: -14.086 mm
y: 66.319 mm
z: 56.500 mm
T1: 0.72580 s
T2: 0.52440 s
T2s: 0.52440 s
ρ: 1.0000
Δw: 0.0000 Hz
0.1
z 0
−0.1
−0.1
y 0
0.1
0
x
0
0.5
1
1.5
2
T1 (s)
B Training randomisation, T1-plate slab
50 mm
shift (mm)
rotation
(1.8, −2.2)
−3.5°
(−5.5, 0.4)
−0.7°
(−3.7, 0.4)
−1.5°
C IR-SE shot
chosen by the policy
TI 0.01-3 s
TE 20 ms (fixed)
180°
90°
180°
180°
RF
Gx
Gy
ADC
TR 0.5-5 s (×32 phase-encode shots per block)
D Fidelity ladder, 14-sphere plate, CPU
Bloch spins per step
40k
20k
0
sphere spins
water spins
cost per RL step
Cost per RL step (s)
1
0.5
0
0.034 s
0.34 s
≈0.34 s
0.42 s
1.03 s
4,816
4,816
6,919
23,735
analytic (closed form)
cached 3 mm
cached 1 mm
full 3 mm
full 1 mm